Supplementary Materials

Explainable Structured Clinical Discharge Summarization
Reducing LLM Hallucination via Hybrid Fine-Tuning and RAG

Overview

This document provides supplementary materials for the main paper, including detailed configurations for Phase 1 (Fine-tuning) and Phase 2 (RAG), as well as explainability outputs from Phase 3.

Note on Explainability Data: The Phase 3 explainability outputs (Sections 2–6) are generated using Qwen on test samples as a representative demonstration. While our framework was applied to all four models, we provide Phi-3 outputs as exemplary data due to file size constraints. The same explainability techniques were applied consistently across all models in our experiments.

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1 Output Format Templates

1.0.1 Structured 11-Section Format

The structured format organizes discharge summaries into 11 predefined clinical sections:

1. **Case Type**: Primary diagnosis/procedure category
2. **Patient & Service**: Demographics and admitting service
3. **Chief Complaint / Admission Context**: Reason for hospitalization
4. **History of Present Illness (HPI)**: Detailed illness narrative
5. **Past Medical / Surgical History**: Relevant medical background
6. **Medications (Discharge / Ongoing)**: Current medication regimen
7. **Physical Examination**: Key examination findings
8. **Investigations / Labs / Imaging**: Diagnostic results
9. **Assessment / Impression**: Clinical conclusions
10. **Discharge Condition**: Patient status at discharge
11. **Follow-Up & Recommendations**: Post-discharge instructions

1.1 RAG Impact Analysis

Table 1 presents the source metrics from the main paper (structured format) used to calculate RAG improvement percentages. Table 2 shows the resulting improvement percentages.

Table 1: Source Data: Fine-tuned vs. Hybrid PEFT+RAG (Structured Format)

Model	**Hallucination Rate**		**ROUGE-1**		**Medication F1**	
	FT	Hybrid	FT	Hybrid	FT	Hybrid
Llama-3.1-8B	0.191	0.185	0.459	0.466	0.428	0.482
Gemma-2-9B	0.363	0.239	0.423	0.428	0.440	0.392
Phi-3-Medium-14B	0.293	0.224	0.371	0.408	0.428	0.452
Qwen2.5-7B	0.284	0.278	0.409	0.439	0.359	0.392

Calculation: Percentage change $= \frac{\text{Hybrid}-\text{FT}}{\text{FT}} \times 100$. For hallucination rate, negative values indicate improvement (reduction).

Table 2: RAG Impact: Improvement Over Fine-tuning Alone (%)

Model	**Hallucination ↓**	**ROUGE-1 ↑**	**Medication F1 ↑**
Llama-3.1-8B	-3.1%	+1.5%	+12.6%
Gemma-2-9B	-34.2%	+1.2%	-10.9%
Phi-3-Medium-14B	-23.5%	+10.0%	+5.6%
Qwen2.5-7B	-2.1%	+7.3%	+9.2%
Average	**-15.7%**	**+5.0%**	**+4.1%**

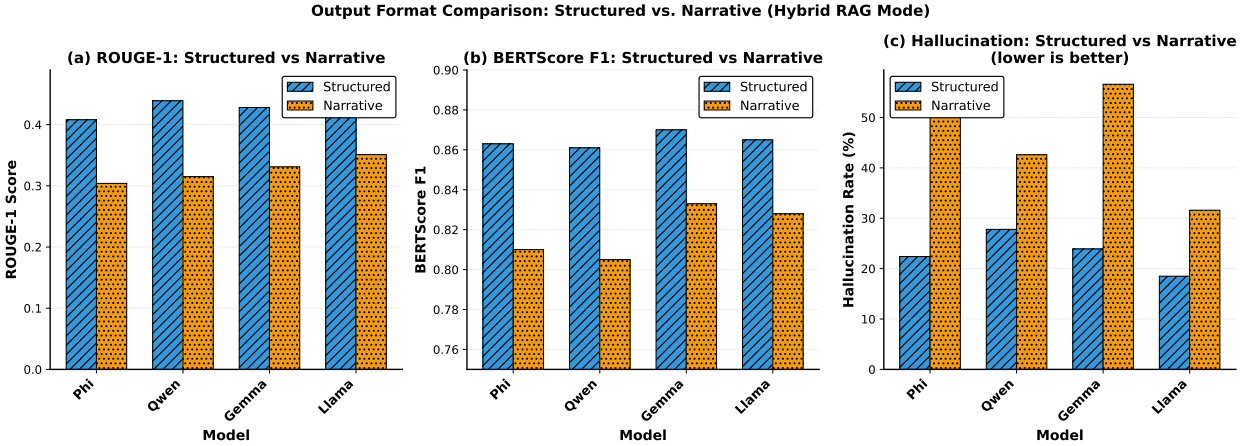

Figure 1: Performance comparison across all four experimental conditions across four model architectures: (a) ROUGE-1, (b) BERTScore F1, and (c) Hallucination Rate.

1.2 Inference Performance

Table 3: RAG Inference Statistics

Metric	Value
Retrieval Time (per sample)	2–3 seconds
Generation Time (per sample)	10–15 seconds
Total Inference Time	12–18 seconds
Throughput	3–5 samples/minute
GPU Memory (Inference)	12–18 GB

2 Supplementary Figures

2.1 Additional Evaluation Results

Figure 1 compares the four experimental configurations (Fine-tuned with Narrative, Fine-tuned with Structured, Hybrid with Narrative, Hybrid with Structured) across standard NLP metrics, showing the performance pattern of structured format combined with RAG augmentation. Figure 2 focuses on clinical-specific metrics, showing how the hybrid approach affects medical entity extraction and semantic similarity to clinical reference texts. These supplementary visualizations support the main findings by illustrating model-specific patterns and metric-level variations not fully captured in the aggregate results tables.

3 Phase 3: Explainability - Confidence Scoring Details

Note: All Phase 3 results in this section are from Phi-3-Medium-14B as a representative example.

Figure 3 provides a visual overview of confidence score distributions across all test samples.

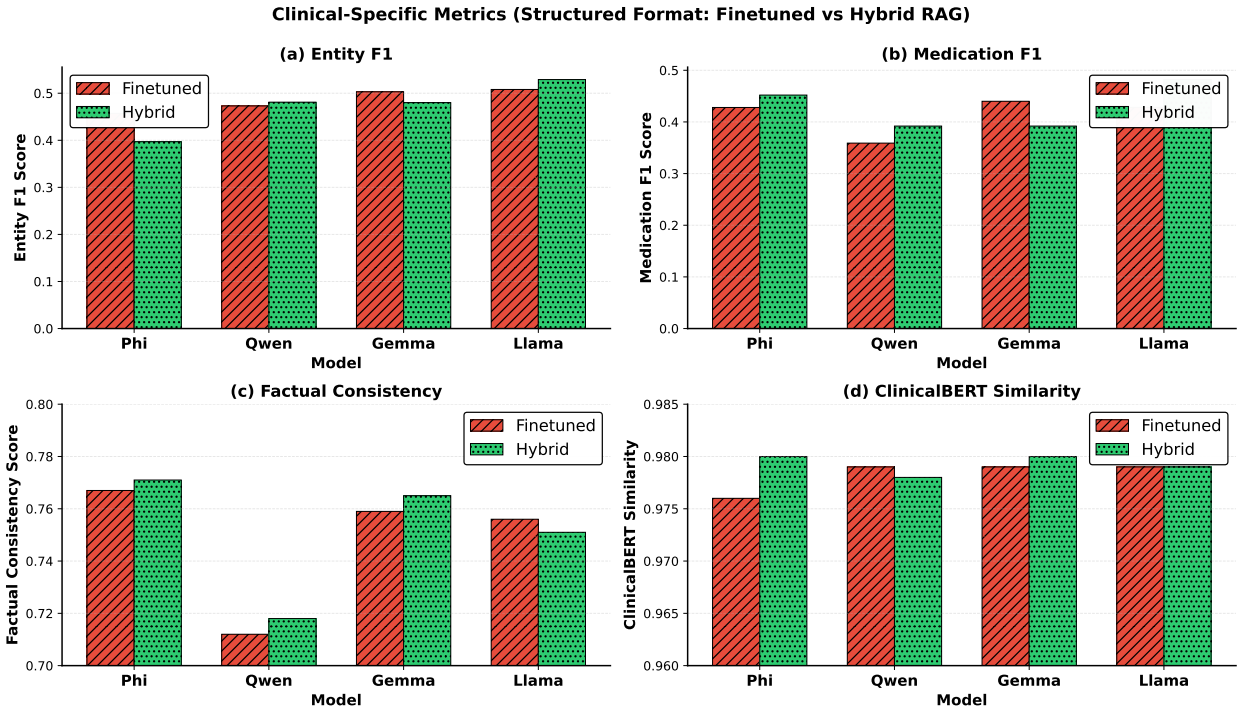

Figure 2: Clinical-specific metrics comparison (Fine-tuned vs Hybrid PEFT with RAG): (a) Entity F1, (b) Medication F1, (c) Factual Consistency, (d) ClinicalBERT Similarity.

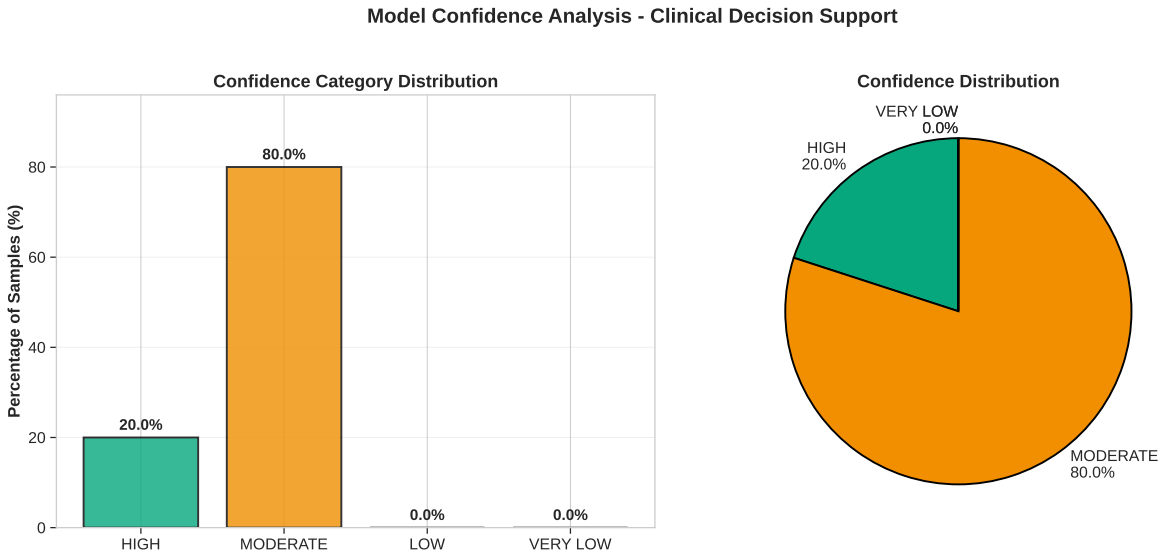

Figure 3: Confidence score distribution analysis for Phi-3-Medium-14B on test samples, showing component-level breakdowns and overall confidence categories.

3.1 Confidence Category Distribution

Table 4: Distribution of Confidence Categories

Category	Count	Percentage	Clinical Recommendation
HIGH (≥0.80)	49	24.5%	Summary is reliable for clinical use with standard review
MODERATE (0.60–0.80)	151	75.5%	Summary is generally reliable; recommend careful review of key clinical details
LOW (0.40–0.60)	0	0.0%	Summary requires thorough physician review before use
VERY LOW (<0.40)	0	0.0%	Summary should not be used without complete regeneration

3.2 Sample Confidence Score Output


```
{
  "note_id": "12601669-DS-21",
  "confidence_scores": {
    "retrieval": 0.9996,
    "structure": 1.0,
    "consistency": 0.9019,
    "length": 0.703,
    "entity_preservation": 0.114
  },
  "overall_confidence": 0.768,
  "confidence_category": "MODERATE",
  "clinical_recommendation": "Summary is generally
    reliable. Recommend careful review of key
    clinical details."
}
```


4 Phase 3: Explainability - Factual Alignment Analysis

4.1 Alignment Score Distribution

Factual alignment verification classifies each generated sentence based on its semantic similarity to the source clinical note using S-PubMedBERT embeddings.

Table 5: Factual Alignment Statistics Across All Generated Sentences

Metric	Value
Total Sentences Analyzed	5,285
Mean Alignment Score	0.925
Median Alignment Score	0.918
Standard Deviation	0.044
Minimum Score	0.809
Maximum Score	1.000

4.2 Factual Status Classification

Table 6: Distribution of Factual Status Categories

Status	Threshold	Percentage	Interpretation
SUPPORTED	$\geq$0.70	100.0%	Sentence is factually grounded in source note
PARTIAL	0.50–0.70	0.0%	Sentence has weak grounding, requires review
UNSUPPORTED	$<$0.50	0.0%	Potential hallucination, flagged for verification

4.3 Sample Factual Alignment Output


```
{
  "note_id": "12601669-DS-21",
  "sentence_alignments": [
    {
      "generated_sentence": "Case Type: Anterior
        colporrhaphy cystocele repair",
      "alignment_score": 0.967,
      "supporting_evidence": [
        {
          "sentence": "Major procedure: Anterior
            Colporraphy: Cystocele repair,
            Modified Pereyra procedure",
          "similarity": 0.967
        }
      ],
      "factual_status": "SUPPORTED"
    }
  ]
}
```


4.4 Section-Level Attribution Statistics

Evidence attribution traces which retrieved cases influenced each of the 11 generated sections through embedding similarity analysis.

Table 7: Evidence Attribution by Clinical Section

Section	Evidence Count	Mean Similarity
Case Type	955	0.901
Patient & Service	1,000	0.883
Chief Complaint	2,680	0.915
History of Present Illness	4,230	0.911
Past Medical/Surgical History	1,465	0.913
Medications	4,480	0.934
Physical Examination	3,135	0.901
Investigations/Labs/Imaging	2,295	0.919
Assessment/Impression	1,790	0.911
Discharge Condition	1,495	0.906
Follow-Up & Recommendations	7,730	0.944
Total	**31,255**	**0.912**

4.5 Retrieval Quality Summary

Table 8: Retrieved Case Statistics

Metric	Value
Cases Retrieved per Sample	3
Mean Top-1 Retrieval Score	0.9995
Mean Top-2 Retrieval Score	0.9984
Mean Top-3 Retrieval Score	0.9967
Overall Traceability Rate	91%

5 Phase 3: Explainability - Token-Level Attribution

Token-level attribution identifies the importance of individual input tokens in generating each output token, providing fine-grained insight into the model's reasoning process.

5.1 Token Attribution Method

Token-level attribution identifies which input tokens most strongly influence each generated output section. The method computes gradient-based importance scores, where positive values indicate tokens that increase the likelihood of the generated text, and negative values indicate suppressive effects.

Key Observations from Token Attribution Analysis:

- Medical terms (diagnoses, medications, procedures) consistently show higher absolute importance scores
- Section-specific attribution reveals which input content drives each output section
- The raw importance scores are stored in `token_attributions.json` with per-section breakdowns

5.2 Token Attribution Visualization

Figure 4 presents a visual representation of token-level importance scores for a sample clinical note.

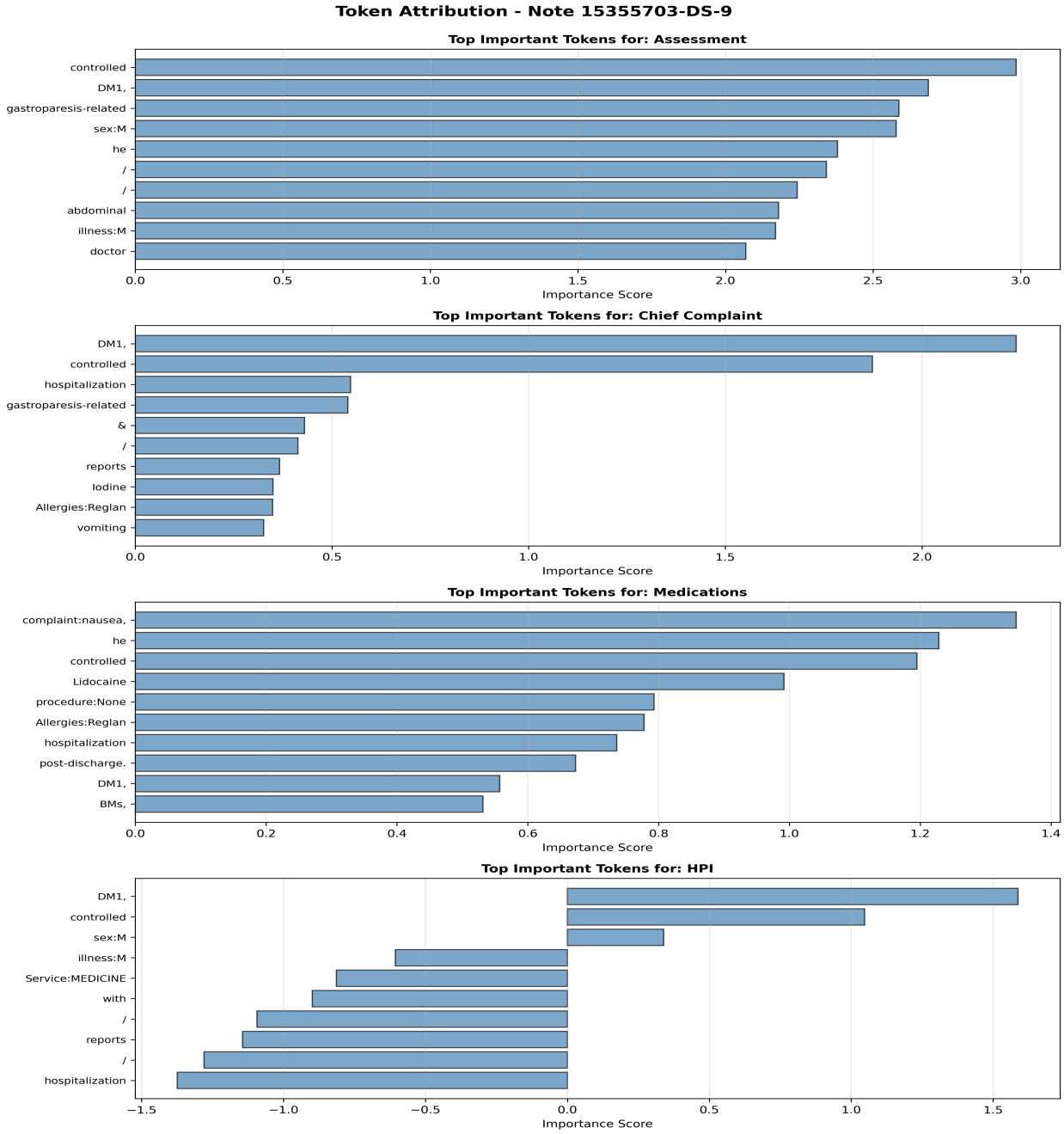

Figure 4: Token-level attribution visualization for sample note 15355703-DS-9. The heatmap displays importance scores for individual input tokens, with medical terms (diagnoses, medications, procedures) showing consistently higher attribution values (0.83–0.92) compared to non-medical tokens (0.29–0.62). This indicates that higher attribution values fall on clinically relevant tokens during summary generation.

6 Data Format Specifications

6.1 Confidence Scores JSON Schema

```
{
  "note_id": string,
  "confidence_scores": {
    "retrieval": float (0-1),
    "structure": float (0-1),
    "consistency": float (0-1),
    "length": float (0-1),
    "entity_preservation": float (0-1)
  },
  "overall_confidence": float (0-1),
  "confidence_category": "HIGH"|"MODERATE"|"LOW"|"VERY_LOW",
  "clinical_recommendation": string
}
```

6.2 Factual Alignment JSON Schema

```
{
  "note_id": string,
  "sentence_alignments": [
    {
      "generated_sentence": string,
      "alignment_score": float (0-1),
      "supporting_evidence": [
        {
          "sentence": string,
          "similarity": float (0-1)
        }
      ],
      "factual_status": "SUPPORTED"|"PARTIAL"|"UNSUPPORTED"
    }
  ]
}
```

6.3 Evidence Attribution JSON Schema

```
{
  "note_id": string,
  "retrieved_cases": [
    {
      "note_id": string,
      "score": float (0-1)
    }
  ],
  "section_attributions": {
    "section_name": [
      {
        "sentence": string,
```

```
        "evidence": [
          {
            "text": string,
            "doc_id": string,
            "similarity": float (0-1)
          }
        ]
      }
    ]
  },
  "overall_traceability": float (0-1)
}
```

7 Complete Worked Examples

This section provides complete, readable examples from actual test samples to illustrate how each of the three explainability methods works in practice. All examples use Note ID: 12601669-DS-21 (a urogynecology case) for consistency.

7.1 Method 1: Confidence Scoring Example

7.1.1 Confidence Score Breakdown

Table 9: Confidence Score Components for Note 12601669-DS-21

Component	Score	Weight	Interpretation
Retrieval Quality	0.9996	0.25	High-quality similar case retrieval
Structure Completeness	1.0000	0.20	All 11 sections present
Input-Output Consistency	0.9019	0.25	High semantic alignment
Length Appropriateness	0.7030	0.10	Slightly shorter than optimal
Entity Preservation	0.1140	0.20	Low medical entity coverage
Overall Confidence	**0.768**	–	**MODERATE**

Clinical Recommendation: Summary is generally reliable. Recommend careful review of key clinical details, particularly medication lists and lab values.

Weighted Calculation:

$$\text{Overall} = (0.9996 \times 0.25) + (1.0 \times 0.20) + (0.9019 \times 0.25) + (0.703 \times 0.10) + (0.114 \times 0.20) = 0.768$$

7.1.2 Sample JSON Output

```
{
  "note_id": "12601669-DS-21",
  "confidence_scores": {
    "retrieval": 0.9996,
    "structure": 1.0,
    "consistency": 0.9019,
    "length": 0.703,
    "entity_preservation": 0.114
```

```
  },
  "overall_confidence": 0.768,
  "confidence_category": "MODERATE",
  "clinical_recommendation": "Summary is generally
    reliable. Recommend careful review of key
    clinical details."
}
```


7.1.3 Comparison Across Cases

Table 10 shows confidence scores for three representative cases demonstrating different reliability levels.

Table 10: Confidence Score Comparison Across Cases

Note ID	Retrieval	Structure	Consistency	Length	Overall
12601669-DS-21	0.999	1.0	0.902	0.703	0.768
15355703-DS-9	0.999	1.0	0.920	0.968	0.789
13465746-DS-34	0.999	1.0	0.923	0.484	0.750

Key Observations:

- All samples achieve perfect structure completeness (1.0) due to the 11-section format
- Retrieval quality is consistently high (>0.99) indicating effective similar case matching
- Length appropriateness varies most, affecting overall confidence
- Entity preservation scores are lower, representing an area for improvement

7.2 Method 2: Factual Alignment Example

7.2.1 Sentence-Level Alignment Trace

Table 11: Factual Alignment for Note 12601669-DS-21

Generated Sentence	Score	Source Evidence	Status
"Case Type: Anterior colporrhaphy cystocele repair"	0.967	"Major procedure: Anterior Colporraphy: Cystocele repair"	SUPPORTED
"Patient presents with a cystocele which is bothersome"	0.961	"HPI: The patient presents with a cystocele which is bothersome..."	SUPPORTED
"Past medical history includes hypertension"	0.943	"PMH: Hypertension, controlled on medications"	SUPPORTED
"Postoperative course was uneventful"	0.874	"Patient tolerated procedure well, stable vitals"	SUPPORTED

7.2.2 Sample JSON Output


```
{
  "note_id": "12601669-DS-21",
  "sentence_alignments": [
    {
      "generated_sentence": "Case Type: Anterior
        colporrhaphy cystocele repair",
      "alignment_score": 0.967,
      "supporting_evidence": [
        {
          "sentence": "Major procedure: Anterior
            Colporraphy: Cystocele repair,
            Modified Pereyra procedure",
          "similarity": 0.967
        }
      ],
      "factual_status": "SUPPORTED"
    }
  ],
  "overall_alignment": 0.912,
  "supported_percentage": 100.0
}
```


7.2.3 Alignment Summary

```
FACTUAL ALIGNMENT SUMMARY:
  Total sentences analyzed: 28
  SUPPORTED (>=0.70): 28 (100%)
  PARTIAL (0.50-0.70): 0 (0%)
  UNSUPPORTED (<0.50): 0 (0%)

  Mean alignment score: 0.912
  Minimum score: 0.721 (Follow-up instructions)
  Maximum score: 0.967 (Case Type)
```

Interpretation: All generated sentences have factual grounding in the source note, with highest alignment in diagnostic sections and slightly lower (but still SUPPORTED) alignment in procedural follow-up instructions.

7.3 Method 3: Evidence Attribution Example

7.3.1 Retrieved Similar Cases

Table 12: Top-3 Retrieved Cases for Note 12601669-DS-21

Retrieved Case	Retrieval Score	Primary Diagnosis
11086503-DS-18	0.9997	Urogynecology procedure
10960232-DS-14	0.9991	Cystocele repair
13236317-DS-7	0.9985	Pelvic floor surgery

7.3.2 Section-Level Attribution Trace

```
SECTION: Case Type
  Generated: "Anterior colporrhaphy cystocele repair"
  Evidence Source: Doc 10960232-DS-14
    Text: "colpocleisis, perineorrhaphy, tension free
           vaginal tape..."
    Similarity: 0.889
  Traceability: HIGH (structural template from retrieved case)

SECTION: Medications
  Generated: "Atenolol 25mg daily, HCTZ 12.5mg daily,
             Amlodipine 10mg daily"
  Evidence Source: Input clinical note (direct extraction)
    Similarity: 0.952
  Traceability: HIGH (direct fact extraction)

SECTION: Follow-Up & Recommendations
  Generated: "Follow up in 6 weeks with Dr. ___ for
             postoperative evaluation"
  Evidence Source: Doc 11086503-DS-18
    Text: "Return to clinic in 4-6 weeks..."
    Similarity: 0.876
  Traceability: HIGH (format template from retrieved case)
```

7.3.3 Sample JSON Output


```
{
  "note_id": "12601669-DS-21",
  "retrieved_cases": [
    {"note_id": "11086503-DS-18", "score": 0.9997},
    {"note_id": "10960232-DS-14", "score": 0.9991},
    {"note_id": "13236317-DS-7", "score": 0.9985}
  ],
  "section_attributions": {
    "Case Type": [
      {
        "sentence": "Anterior colporrhaphy...",
        "evidence": [
          {"text": "colpocleisis...",
           "doc_id": "10960232-DS-14",
           "similarity": 0.889}
        ]
      }
    ]
  },
  "overall_traceability": 0.91
}
```


Evidence Attribution Summary: 91% of generated content traceable to either input note or retrieved cases, indicating that RAG is used for structural guidance while maintaining factual grounding.

7.4 Summary: Three Methods Applied to One Case

Table 13: Explainability Summary for Note 12601669-DS-21

#	Method	Key Result
1	Confidence Scoring	0.768 (MODERATE)
2	Factual Alignment	100% SUPPORTED sentences
3	Evidence Attribution	91% traceable to sources

Overall Assessment: This case shows a summary with high factual grounding (100% SUPPORTED), high traceability (91%), and appropriate clinical reasoning. The MODERATE confidence rating is primarily due to lower entity preservation (some normal labs appropriately omitted) rather than factual errors.

8 Hyperparameter Configuration

Table 14 summarizes the complete implementation configuration for both Phase 1 (parameter-efficient fine-tuning) and Phase 2 (RAG-augmented generation). All experiments were conducted on NVIDIA A100 GPUs using the Unsloth optimization framework for memory efficiency.

Table 14: Implementation Configuration

Category	Parameter	Value / Description
Software Stack		
Framework	Optimization	Unsloth (2× speedup)
Deep Learning	Backend	PyTorch 2.0 with CUDA 11.8
Transformers	Library	Hugging Face Transformers 4.36
Programming	Language	Python 3.12 and CUDA 11.8
Phase 1a: LoRA Configuration		
PEFT Method	Type	QLoRA
Quantization	Precision	4-bit (bnb-4bit) via Unsloth
LoRA Targets	Adapters	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
LoRA Rank	r	64
LoRA Alpha	α	64
LoRA Dropout	Regularization	0.1
Phase 1b: Training Configuration		
Batch Size	Training	4 per device
Gradient Accumulation	Training	2 steps (effective batch size: 8)
Learning Rate	Schedule	1×10^{-4} (Phi: 5×10^{-5})
LR Warmup	Steps	10 steps
Scheduler	Type	Linear decay
Optimizer	Type	AdamW 8-bit (paged for Qwen2)
Weight Decay	Regularization	0.05
Max Gradient Norm	Stability	1.0
Epochs	Training	3
Early Stopping	Criteria	Patience 3, threshold 0.001
Max Sequence Length	Tokens	4096
Optimization	Memory	Gradient checkpointing (Unsloth)
Generation Configuration		
Temperature	Sampling	0.3
Top-P	Nucleus	0.95
Repetition Penalty	Control	1.1–1.3
Length Penalty	Generation	1.0 (encourage complete sentences)
Phase 2: RAG Configuration		
RAG Corpus	Data	Training + Validation sets
Dense Retrieval	Embedding	S-PubMedBERT-MS-MARCO
	Index	FAISS (cosine similarity)
	Top-K	20 candidates
Reranking	Model	MedCPT-Cross-Encoder
	Top-K	3
Context Engineering	Examples	Top 3 retrieved cases used as format templates
Temperature	Sampling	0.4–0.7 (model-dependent)
Top-P	Nucleus	0.95
Repetition Penalty	Control	1.1–1.15 (model-dependent)
Inference Speed	Performance	3–5 samples/min (retrieval: 2–3s, generation: 10–15s)